
Embedded System

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Course info

- ❑ IT4210E (3-0-1-6).
- ❑ Wednesday 6:45 AM – 9:10 AM + 3 labs.
- ❑ Pre-requisites
 - | Computer Architecture
 - | Electronics for IT
 - | C programming

Course contents

- ❑ Overview of embedded system organization, including hardware (the MCU, memory, peripherals) and software.
- ❑ Designing embedded system hardware with basic peripheral devices.
- ❑ Develop and simulate embedded software based on assembly and high level languages for basic functions, including peripherals (timer/counter, interrupt...), digital input (switch, button, keypad, ...), digital output (LEDs, 7-seg LEDs, LED matrix,...), analog interface (ADC,DAC), and serial communications (UART, SPI, I2C, USB,...).
- ❑ Introduction to embedded RTOS.

What's the interesting challenge?

- ❑ Embedded system is a very very large field, this course only scratches the surface
- ❑ Enhanced engineering skills
 - | Working with hardware layer
 - | Interfacing with the real physical world
- ❑ Hands-on experience
 - | Learning and developing on development kit

Syllabus (tentative)

- ❑ Chap 1: Introduction to embedded systems
- ❑ Chap 2: Embedded system organization
- ❑ Chap 3: Basic peripheral interfacing
- ❑ Chap 4: Arduino
- ❑ Chap 5: Serial communication
- ❑ Chap 6: Real-world interfacing
- ❑ Chap 7: FreeRTOS

Labs

- ❑ Lab 1: Basic peripheral interfacing
- ❑ Lab 2: Serial communication
- ❑ Lab 3: Embedded OS

Course evaluation

- ❑ Progress evaluation:
 - | Labs
- ❑ Final exam: quiz + project-based
 - | Teams of 2-3 students
 - | Project proposed by team (preferred),
 - | or selected from list.
- ❑ Some samples:
 - | [Syllabus - EE319K \(utexas.edu\)](#)
 - | [EE319K Lab 10 Videos \(utexas.edu\)](#)



References

❑ Books and lecture notes

- | Han-Way Huang, Leo Chartrand 2004, Microcontroller: An Introduction to Software & Hardware Interfacing, Cengage Learning.
- | Embedded Systems syllabus from trường Univ. of Cincinnati (EECE 6017C), Univ. of California, Berkeley (EECS 149), Univ. of Pennsylvania (ESE 350), Univ. of Kansas (EECS388)

❑ Technical documents and tools

- | Datasheet from chip maker: Atmel, Keil...
- | Internet
- | Arduino IDE
- | STM32Cube IDE

Chapter 1: Introduction

Content

- ❑ What is embedded system
- ❑ Embedded application
- ❑ Characteristics of embedded systems

Definition

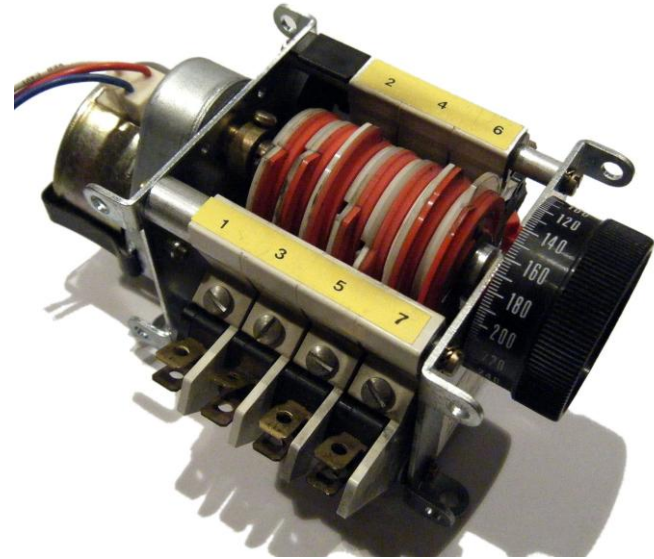
“Dortmund“ Definition: [Peter Marwedel]

Embedded systems are information processing systems embedded into a larger product.

Wikipedia:

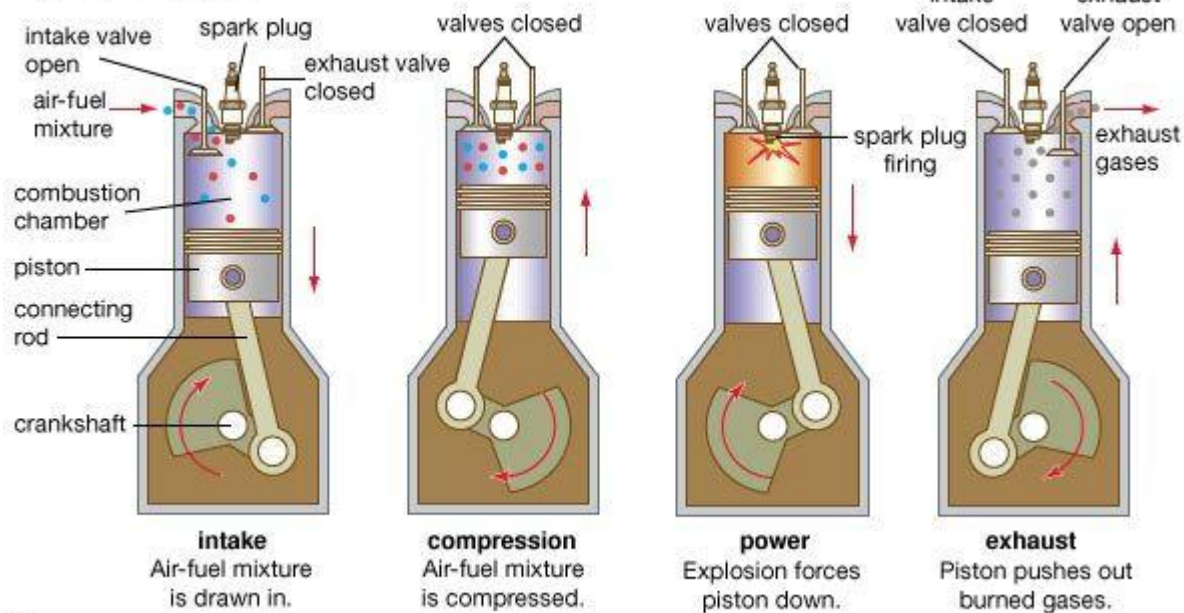
An embedded system is a computer system with a dedicated function within a larger mechanical or electrical system, often with real-time computing constraints. It is embedded as part of a complete device often including hardware and mechanical parts.

A sample



Another sample: ignition in combustion engine

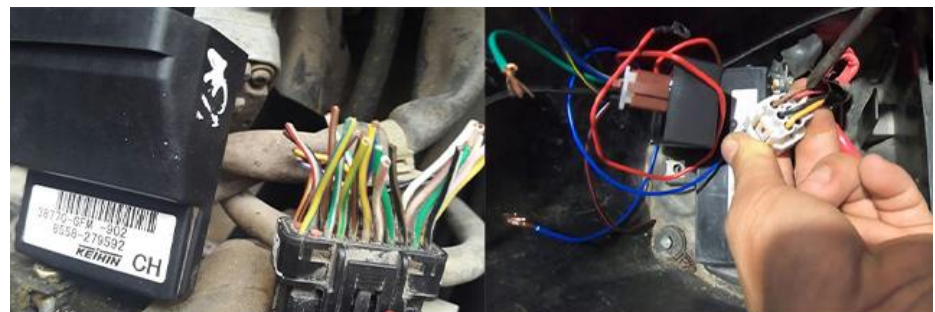
Four-stroke cycle



© 2007 Encyclopædia Britannica, Inc.



Mechanical ignition (distributor)



CỤC IC XE MÁY LÀ GÌ ?
cách phòng chống để xe máy không bị trộm IC

Benefit of embedded system

❑ Before embedded system

- | Logic circuit, hard-wired
- | Electro-mechanical control system
- | Hydraulic control system
- | ...

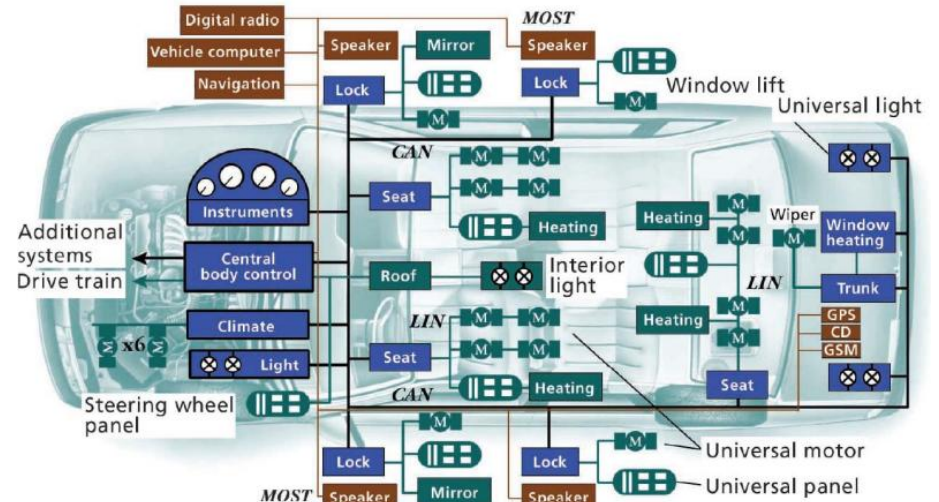
❑ Benefit of embedding and embedded system:

- | sophisticated control, precise timing, low unit cost, low development cost, high flexibility, small size, and low weight

❑ System/final product will have improved

- | Performance
- | Features
- | Cost
- | Dependability

Example



CAN Controller area network
 GPS Global Positioning System
 GSM Global System for Mobile Communications
 LIN Local interconnect network
 MOST Media-oriented systems transport

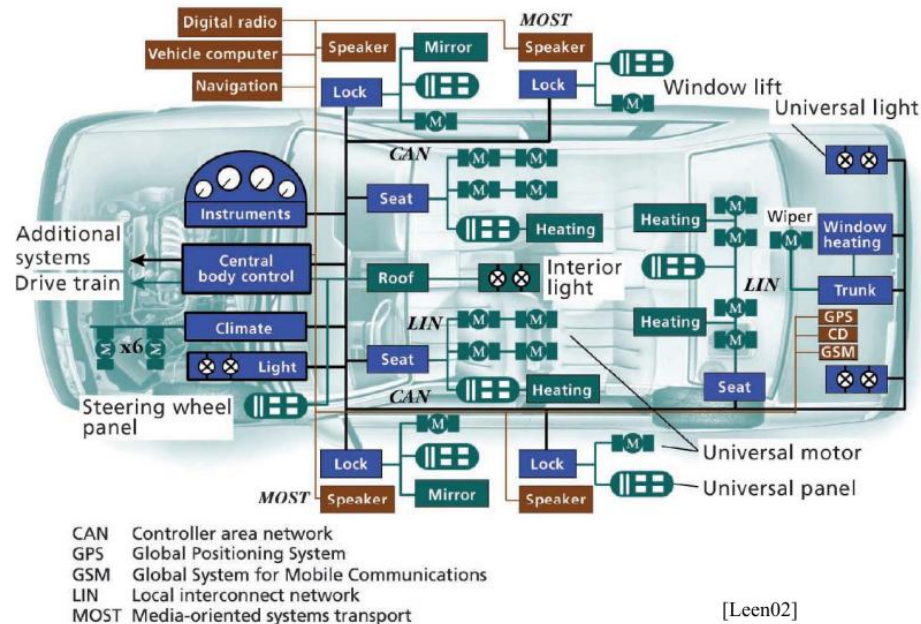
[Leen02]

Automotive electronics

- ABS: Anti-lock braking systems
- ESP: Electronic stability control
- Airbags
- Automatic gearboxes
- Smart keys
- Blind-angle alert systems
- ...

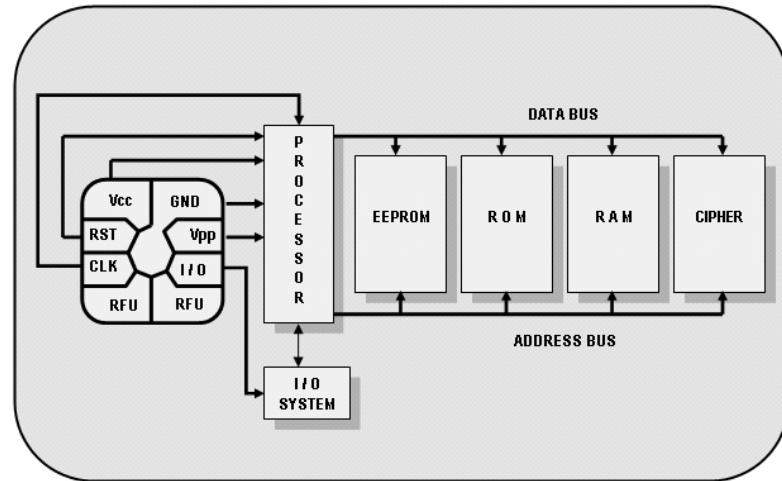
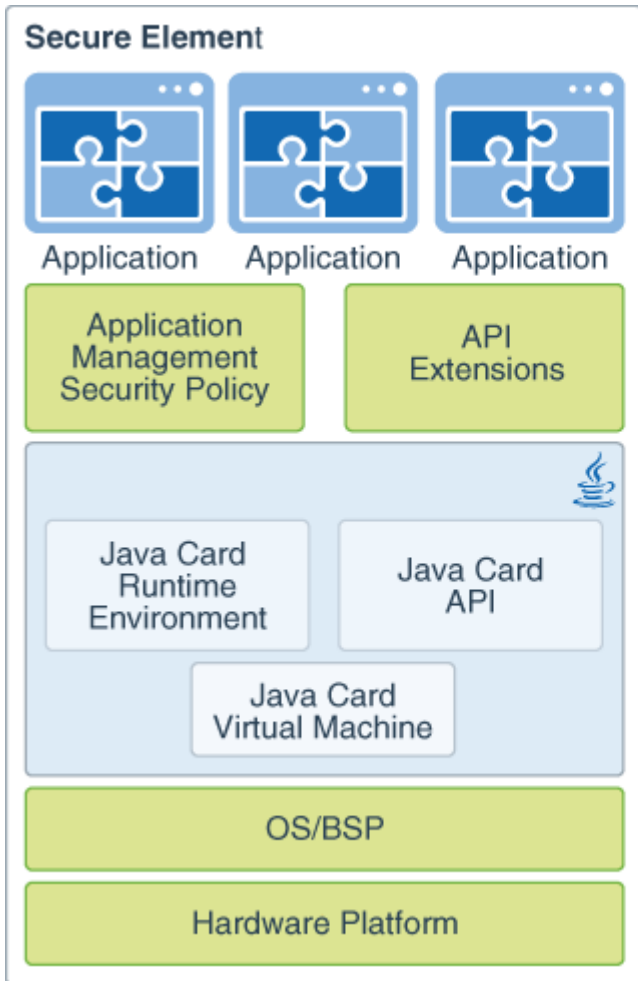


❑ Up to 100 networked processors



[Leen02]

Smart card (vd: SIM card)



Key

- Oracle Java Card
- Product Specific

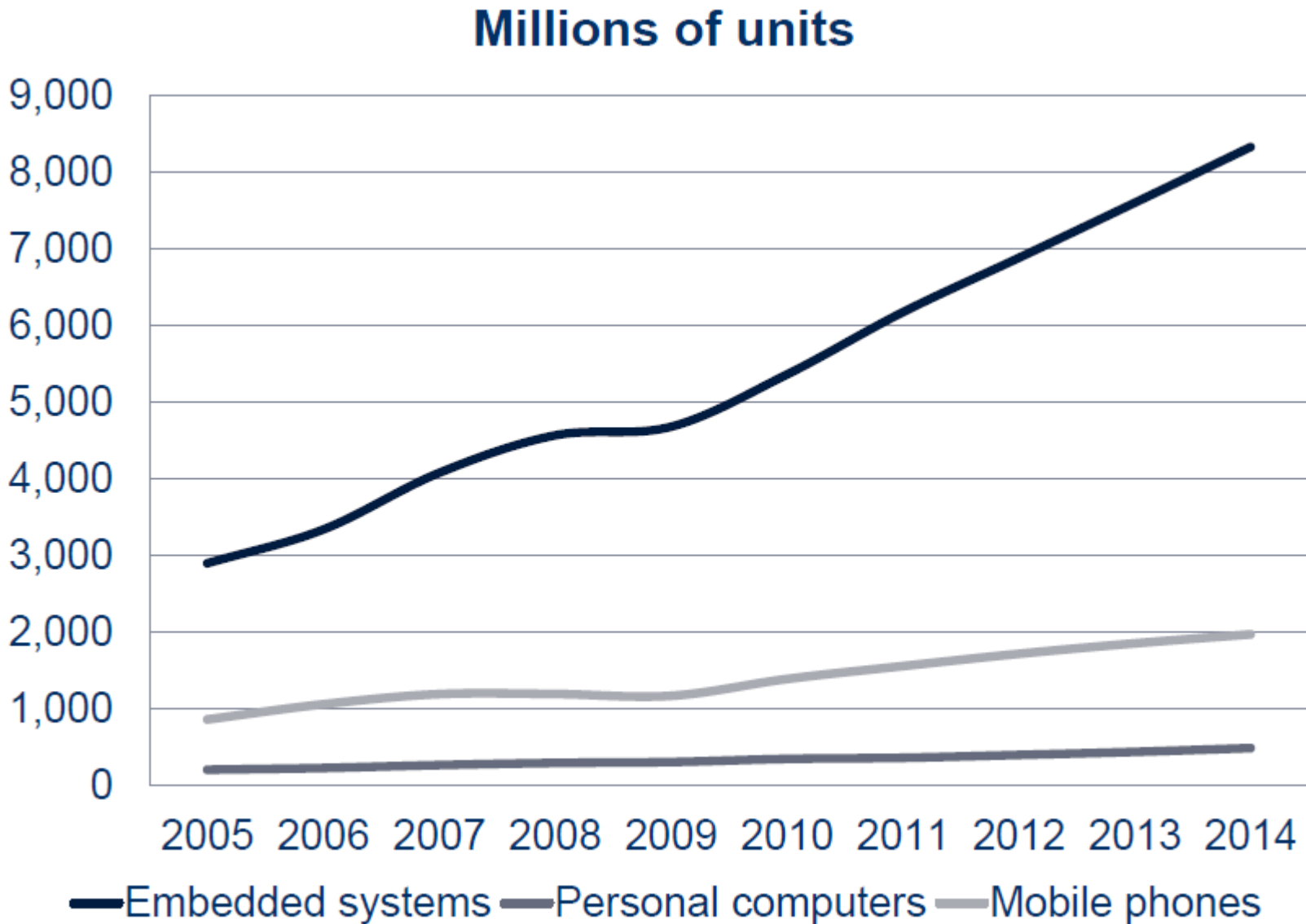
Logistic

- | Object tagging :
RFID, barcode...
- | Object tracking.



Location	Date	Local Time	Activity
TOKYO, JP	09/03/2016	3:48 P.M.	Delivered
Tokyo, Japan	09/02/2016	7:10 P.M.	Transferred to a local agent for delivery. The delivery will be rescheduled.
	09/02/2016	7:09 P.M.	The delivery change was completed. / The package is being held for a future delivery date.
	09/02/2016	10:07 A.M.	A delivery change for this package is in progress. / We've rescheduled this delivery.
Tokyo, Japan	08/30/2016	7:18 P.M.	The receiver was not available at the time of the final delivery attempt. / We'll contact the sender or receiver about this delivery.
	08/30/2016	2:28 P.M.	The receiver was not available at the time of the final delivery attempt.
Tokyo, Japan	08/29/2016	11:17 A.M.	The receiver was not available at the time of the second delivery attempt. A final attempt will be made.
	08/29/2016	7:48 A.M.	The package is awaiting clearing agency review. / Your package was released by the clearing agency.
Tokyo, Japan	08/26/2016	8:30 P.M.	Transferred to a local agent for delivery. The delivery will be rescheduled.
	08/26/2016	2:15 P.M.	The receiver was not available at the time of the first delivery attempt. A second attempt will be made.
	08/26/2016	1:11 P.M.	Destination Scan
	08/26/2016	1:00 P.M.	Arrival Scan
	08/26/2016	12:55 P.M.	Departure Scan
	08/26/2016	12:38 P.M.	Import Scan
Narita, Japan	08/26/2016	9:03 A.M.	Arrival Scan
	08/26/2016	8:44 A.M.	A late flight has caused a delay. We're adjusting plans and working to deliver your package as quickly as possible.
Anchorage, AK, United States	08/25/2016	8:24 A.M.	Departure Scan
	08/25/2016	6:32 A.M.	Arrival Scan
Ontario, CA, United States	08/25/2016	2:21 A.M.	Departure Scan
Ontario, CA, United States	08/24/2016	11:10 P.M.	Arrival Scan
Commerce City, CO, United States	08/24/2016	9:55 P.M.	Departure Scan
Denver, CO, United States	08/24/2016	8:24 P.M.	Arrival Scan
Boulder, CO, United States	08/24/2016	7:35 P.M.	Departure Scan
	08/24/2016	7:21 P.M.	Origin Scan
United States	08/24/2016	5:03 P.M.	Order Processed: Ready for UPS

Market size of embedded system (IDC)



Embedded system market size

Global Embedded System Market Trends



Asia Pacific is expected to witness high growth in the near future as it houses major global semiconductor and electronics companies.

ASIA PACIFIC



e-estimated, p-projected

86.5

USD Billion
2020-e

CAGR of
6.1%

The global embedded system market size is estimated to be USD 86.5 billion in 2020 and projected to reach USD 116.2 billion by 2025, at a CAGR of 6.1%.

116.2

USD Billion
2025-p



The increasing global demand for communication equipment and consumer electronics equipped with embedded systems to drive growth of embedded systems market.



New product developments would offer lucrative opportunities for market players in the next five years.



COVID-19 is expected to adversely impact the demand for consumer electronics and automotive vehicles globally, thus, in turn, hampering the market sales for embedded system and related components.



China is expected to offer lucrative growth opportunities for companies offering embedded components during the forecast period.

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1.2 Embedded systems characteristics

❑ Dependability

- | *Reliability* $R(t)$ = probability of system working correctly provided that it was working at $t=0$
- | *Maintainability* $M(d)$ = probability of system working correctly d time units after error occurred.
- | *Availability* $A(t)$: probability of system working at time t
- | *Safety*: no harm to be caused
- | *Security*: confidential and authentic communication

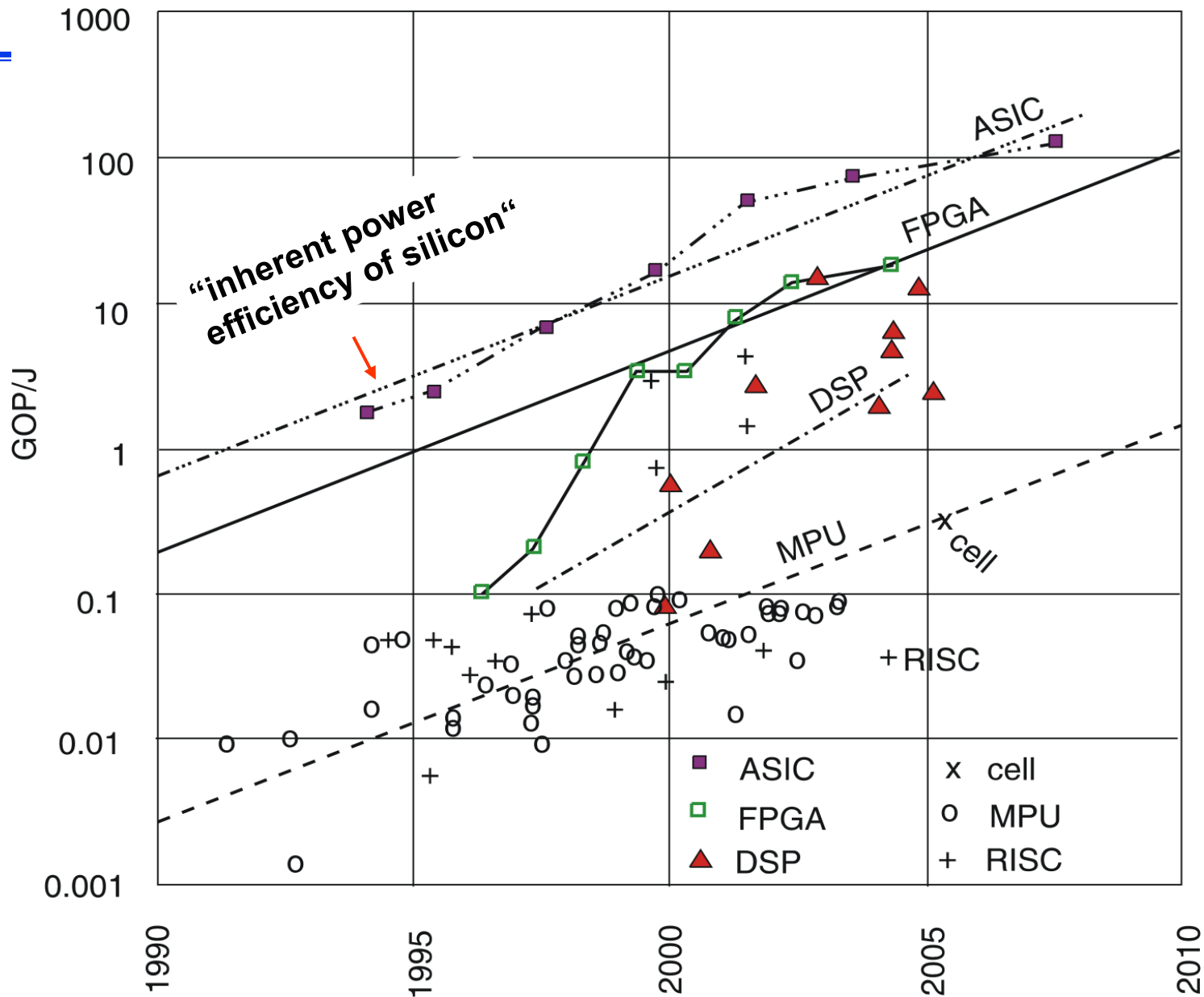
Efficiency and resource constraints

❑ Efficiency

- | Code-size efficient
- | Run-time efficient
- | Weight efficient
- | Cost efficient
- | Energy efficient

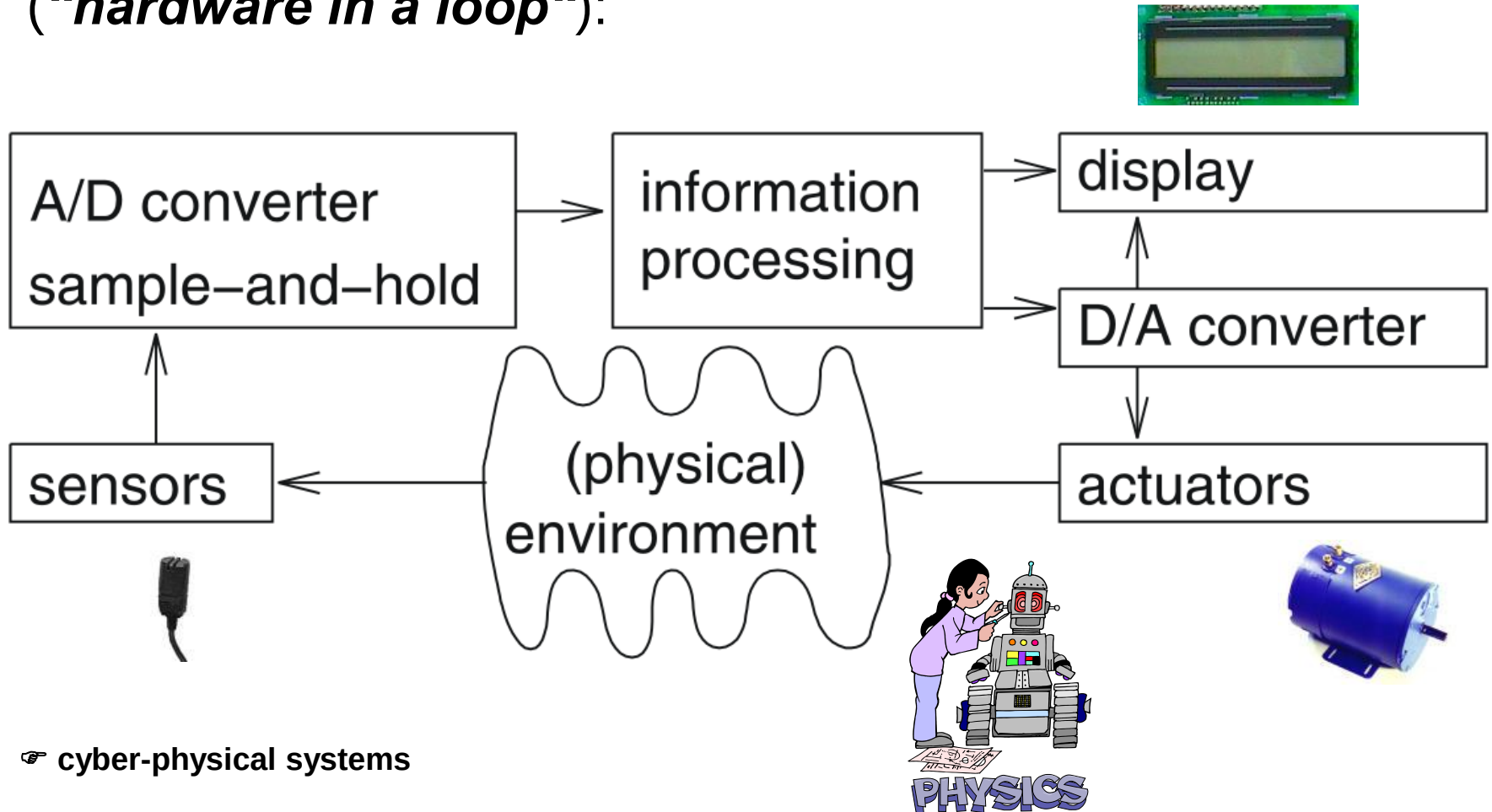
❑ Resource constraints

- | Cost
- | Computing power and storage capacity
- | Physical size
- | Energy consumption



Embedded System Hardware

- ❑ Embedded system hardware is frequently used in a loop (***“hardware in a loop”***):



☞ cyber-physical systems

Responsiveness and Real-time constraints

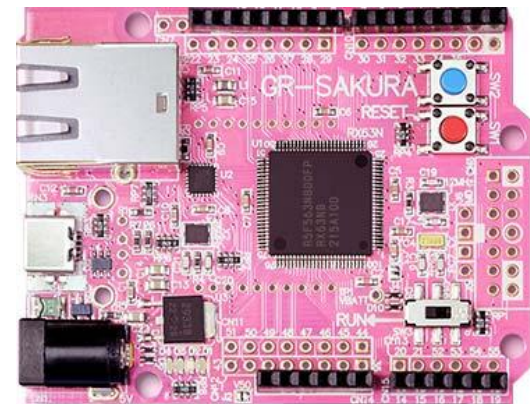
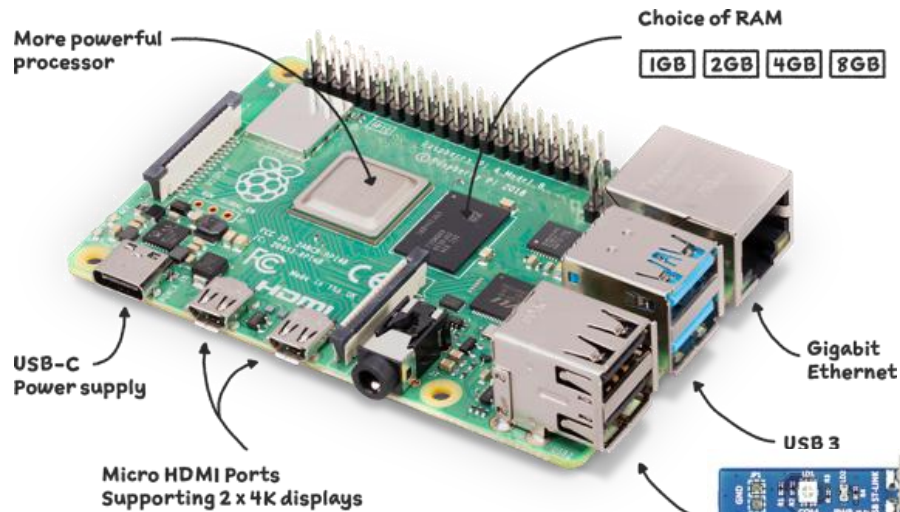
- ❑ React to stimuli from the controlled object or external/internal events within the time allowed.
- ❑ Many ES must meet **real-time constraints**
 - | For real-time systems, right answers arriving too late are wrong.
 - “A real-time constraint is called **hard**, if not meeting that constraint could result in a catastrophe” [Kopetz, 1997].
 - | All other time-constraints are called **soft**.
 - | A guaranteed system response has to be explained without statistical arguments

Dedicated systems for special purpose

- ❑ **Dedicated** towards a certain **application**
Knowledge about behavior at design time can be used to minimize resources and to maximize robustness
- ❑ **Dedicated user interface**
(no mouse, keyboard and screen)

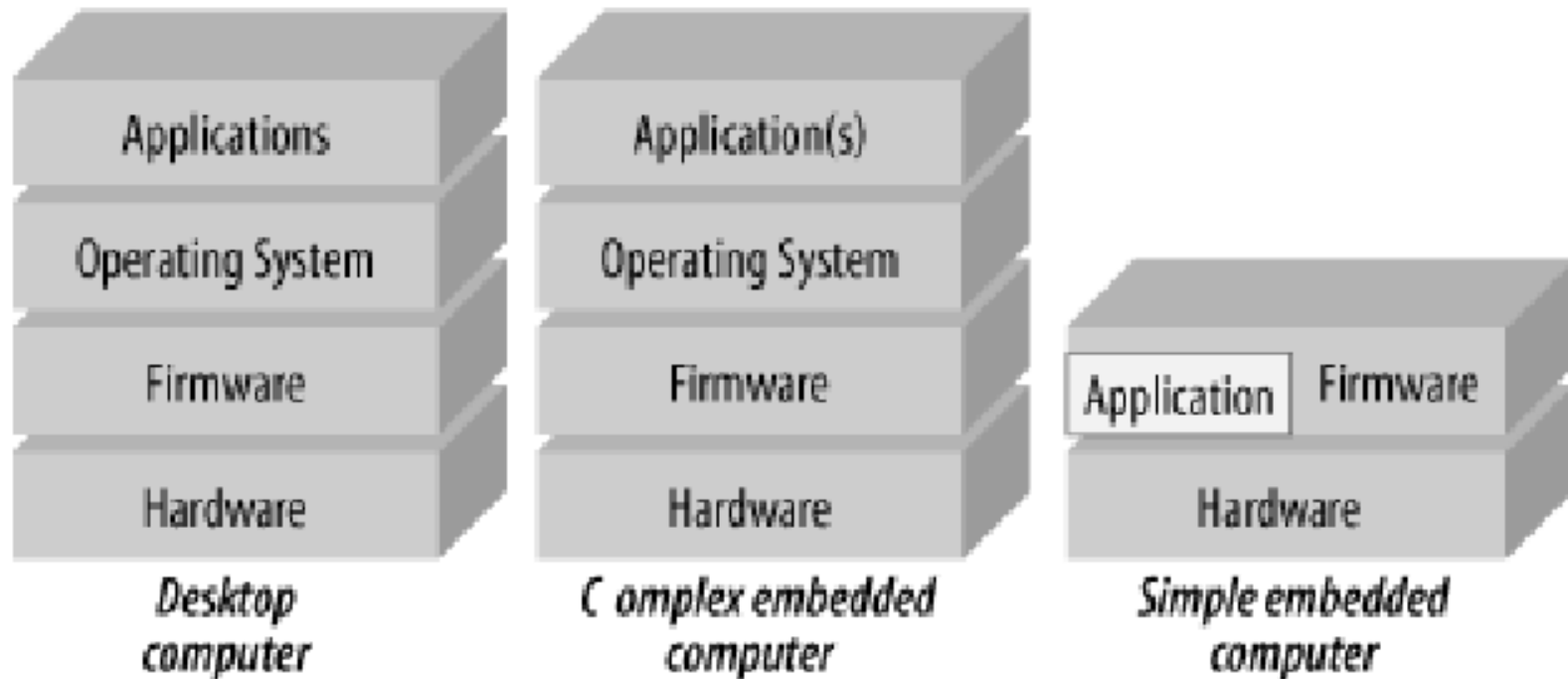


1.3. Embedded computer organization



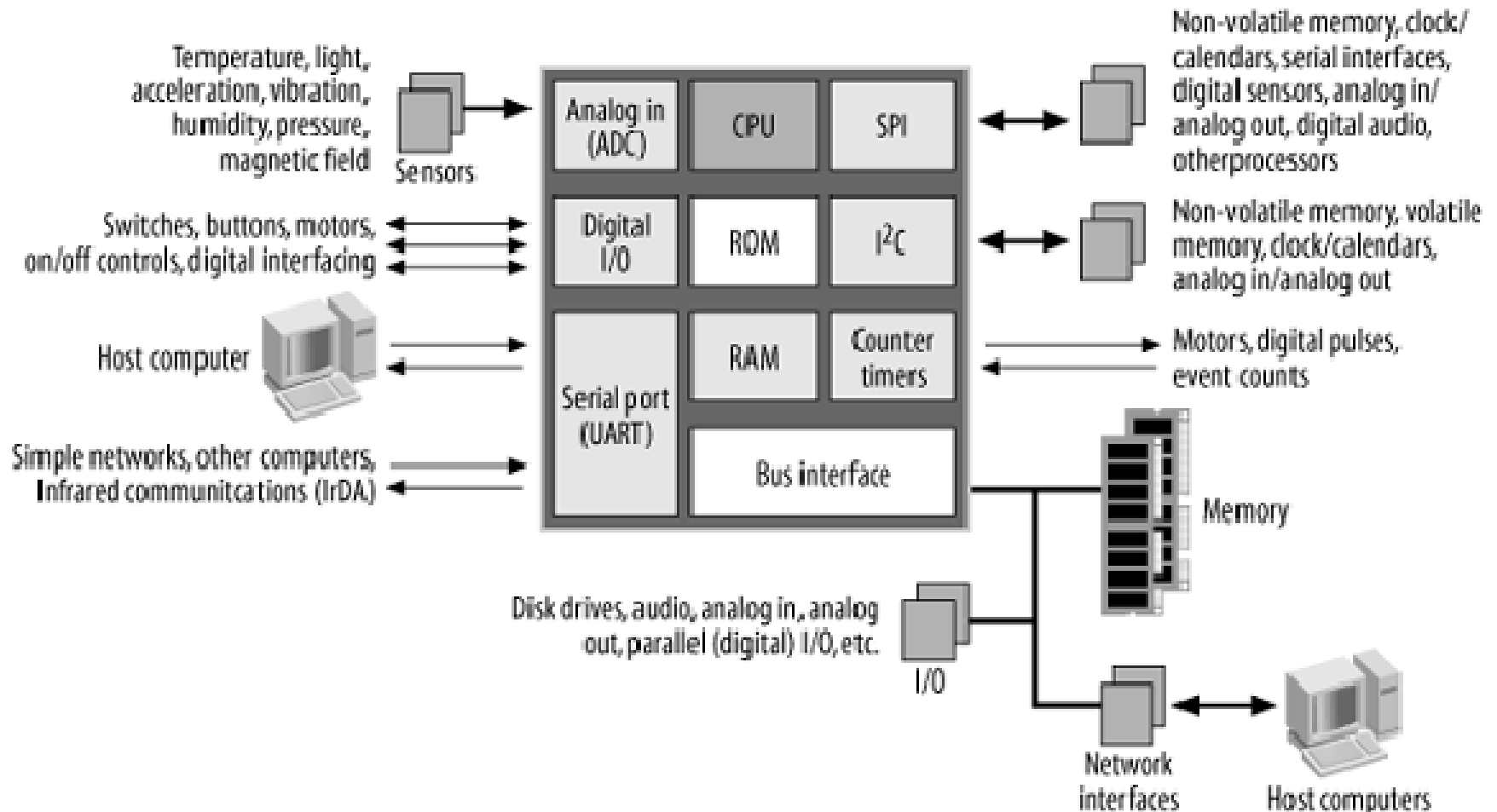
1.3. Embedded computer organization

❑ System structure



Embedded computer organization

❑ Central system organization: MCU



Processing unit in embedded computer

❑ Microprocessor:

- | CPU packaged as a separate IC.
- | Example: Intel Core, AMD Ryzen,...

❑ Microcontroller: (Bộ vi điều khiển)

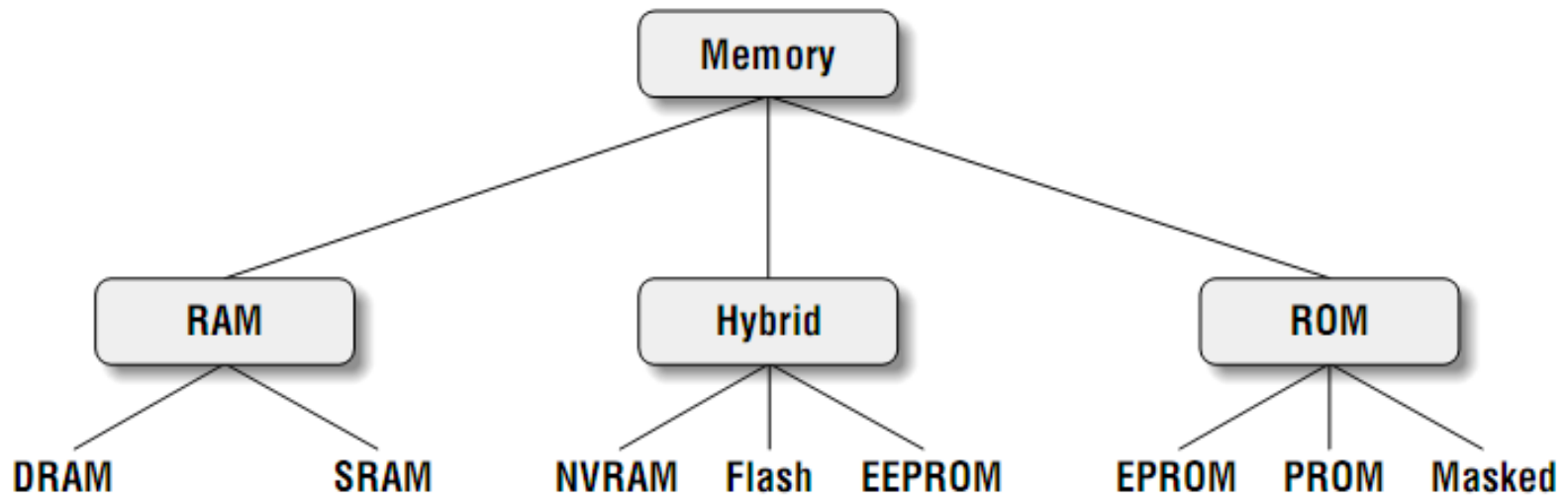
- | A single IC contains basic components of a computer: CPU, Memory (ROM and RAM), basic I/O.
- | Example: AVR from Atmel, PICs from Microchip, STM32F4 (ARM-based) from STMicroelectronics, ...
- | Architecture: 8 bit, 16 bit, 32 bit, (64 bit).
- | On-chip memory or external memory.

❑ SoC (System-on-chip):

- | High performance components on a single IC: CPU, GPU, modem...
- | Example: Samsung Exynos, Qualcomm Snapdragon, Apple M1...

Memory

- ❑ RAM
- ❑ ROM, EPROM, EEROM (EEPROM)
- ❑ Flash: newest ROM technology, market dominating



Input/Output: peripherals

❑ Types of peripherals

- | Input: sensors, button, switch, knob, ADC...
- | Output: display, actuators, DAC...
- | Communication: UART, SPI, I2C, CAN,...
- | Storage: EEPROM, Flash memory,...

❑ I/O programming

- | Programmed I/O (polling)
- | Interrupt-driven I/O
- | Direct Memory Access (DMA)

Exercise

- ❑ Read IC datasheet
 - | 74HC00
 - | DS18B20
- ❑ What is the function of each IC
- ❑ Footprint of each IC
- ❑ How each IC works

End of Chap.1